

Yakub Mungai

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Education

University of Florida

Bachelor of Science in Computer Science

Certificate in AI Fundamentals & Applications

Gainesville, FL

August 2021 – December 2025

Relevant Coursework: Data Structures & Algorithms, Intro to Software Engineering, Algorithm Abstraction and Design, Programming in C++, Programming in Python, Information & Database Systems, Business Analytics & AI.

Technical Skills

- **Languages:** Python, C++, SQL, JavaScript, MATLAB.
- **Frameworks & Libraries:** React Native, Expo, Node.js, Express.js, pandas, NumPy, SciPy, Matplotlib.
- **Databases & Cloud:** MongoDB, Firebase (App/Messaging), REST APIs.
- **Developer Tools:** Git, Visual Studio Code, PyCharm.

Professional Experience

Lead Full-Stack Developer

React Native, Expo, Node.js, Express.js, MongoDB

MacroIQ (Mobile Startup)

Jan 2026 – Present

- Inherited a beta React Native codebase, executing a comprehensive UI/UX refactor to deliver an intuitive, native iOS-aligned aesthetic.
- Redesigned and streamlined the user onboarding workflow, reducing initial friction and optimizing user state initialization.
- Developed and documented RESTful API endpoints using Node.js and Express.js to manage user credentials and secure nutrition logging data.
- Optimized conversational AI co-pilot response latency by 57% (from 4.2s to 1.8s) by designing a bypass architecture that eliminated sequential API round-trips for database mutations.
- Integrated Spike API services alongside Google Gemini models to deliver automated image-based and text-based nutritional analysis from user inputs.

Freelance Software Engineer

SvelteKit, Tailwind CSS, JavaScript, Node.js, Resend API

Self-Employed

Jan 2026 – Present

- **Community Management Platform:** Architected and deployed a full-stack web application for the Tanzania Sharing Association, transforming manual processes for registration, membership renewals, and benefit claims into secure, automated digital workflows.
- **Creative Portfolio Hub:** Programmed a responsive, high-performance web portfolio for digital editor Daniel "DNA" Santos, engineering an optimized serverless pipeline via the Resend API that achieved a 30% increase in inbound client inquiries.

Technical Projects

Project KOSMOS

System Developer (Senior Capstone)

Python, LangChain, OpenAI API, kRPC

August 2025 – December 2025

- Built an LLM critic agent architecture designed to generate natural language evaluations of flight mechanics by consuming real-time execution parameters at sub-second intervals.
- Formulated logic modules to identify programmatic system deviations and dynamically issue automated correctional telemetry overrides.
- Scripted direct integrations with flight simulation software via automated API connection wrappers to continuously observe system states.

AlexG Automator

Quantitative Systems Architect

Python, MT5 API, Pandas, SciPy, Matplotlib

January 2026

- Architected a high-throughput financial data ingestion model leveraging the MetaTrader 5 interface for multi-timeframe analysis.
- Implemented historical and active data filters through Pandas, utilizing structural signal logic to pinpoint localized mathematical market inflection thresholds.
- Constructed low-latency data rendering modules using Matplotlib to map decision criteria trends alongside incoming streaming updates.

Involvements & Leadership

- **Vice President**, ColorStack UF
- **Member**, University of Florida Honors Program
- **Recipient**, UWC Davis Scholarship

May 2024 – Present

August 2021 – December 2025